Smart Exam Planner, Revision Tracker & AI Study Assistant

Project Using Python Tkinter and Excel

NAME: PREETTY. S

REG. NO.: 24MIA1155

DATA MODELLING USING SPREADSHEET

Introduction

Students often struggle to:

- Manage multiple subjects
- Track revision progress
- Identify weak areas

This project provides a **smart desktop application** to:

- Plan study topics
- Track progress
- Analyze performance
- Generate AI-based notes

Objective

- Develop a smart study planner
- Track subject-wise progress
- Identify weak areas automatically
- Provide recommendations
- Integrate AI for note generation
- Use Excel as backend storage

Technologies Used

- **Python** – Core logic
- **Tkinter** – GUI
- **Pandas** – Data handling
- **Matplotlib** – Charts
- **Excel (.xlsx)** – Storage
- **OpenAI API** – AI Assistant

System Architecture

Flow:

User → Tkinter GUI → Python Logic → Pandas → Excel → Output

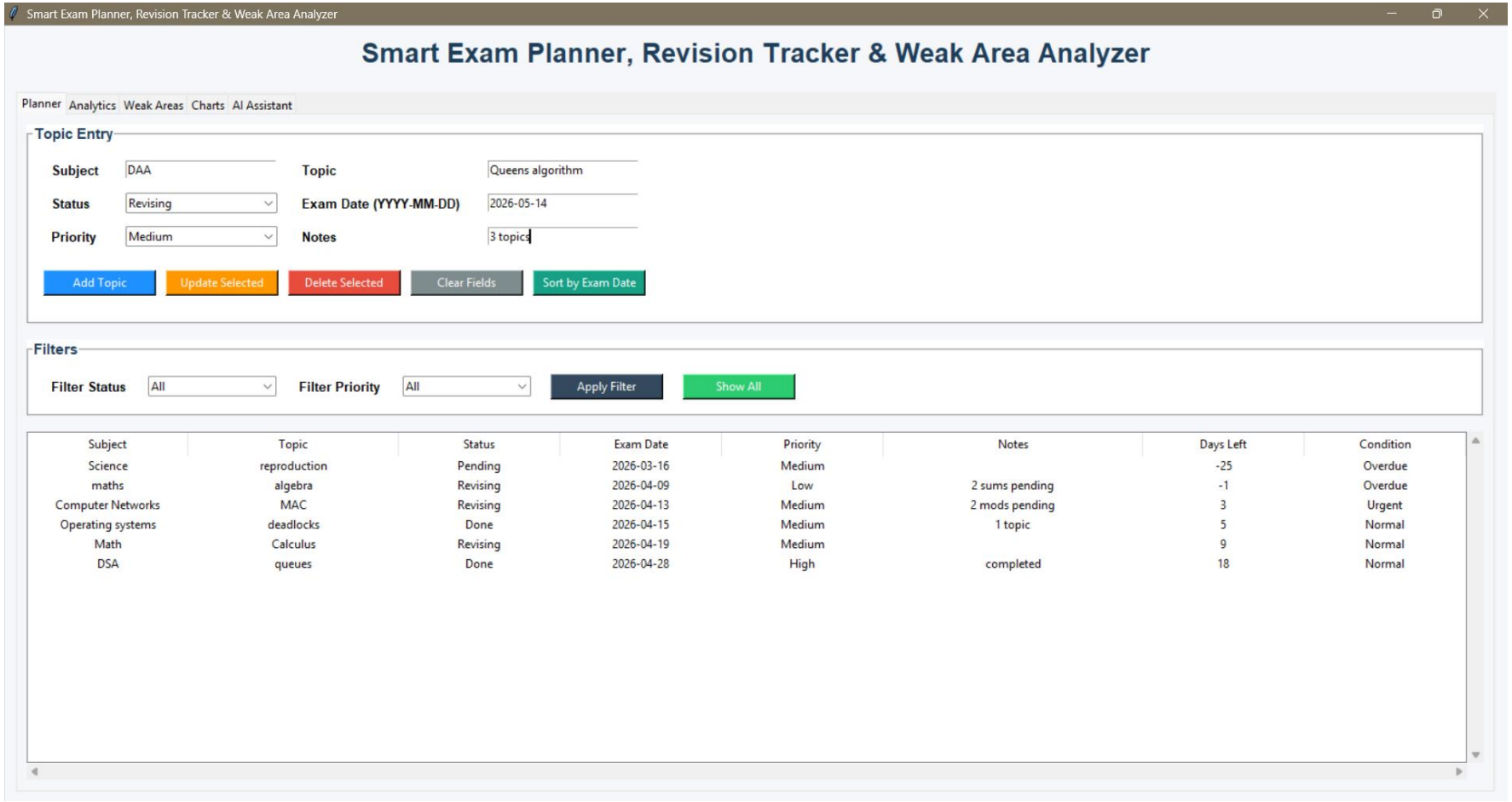
KEY FEATURES

- Add / Update / Delete topics
- Status tracking (Pending / Revising / Done)
- Priority management
- Days left calculation
- Weak area detection
- Recommendation system
- Charts visualization
- AI-based notes generation

DATA MANAGEMENT

- Data stored in Excel
- Managed using Pandas
- Missing values handled (fillna)
- Excel used only as **intermediary storage**

USER INTERFACE – PLANNER TAB

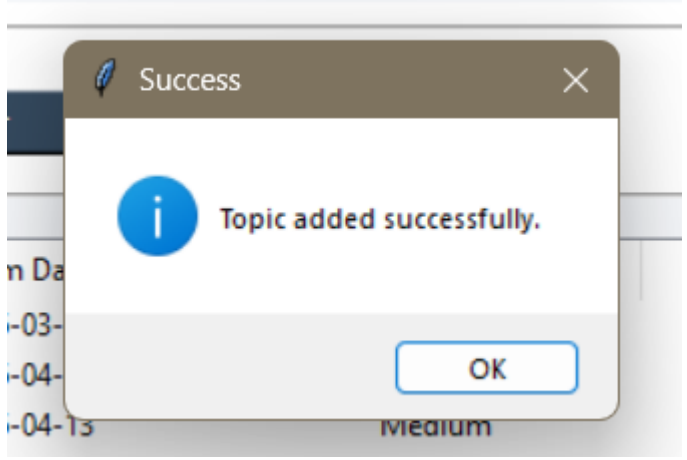


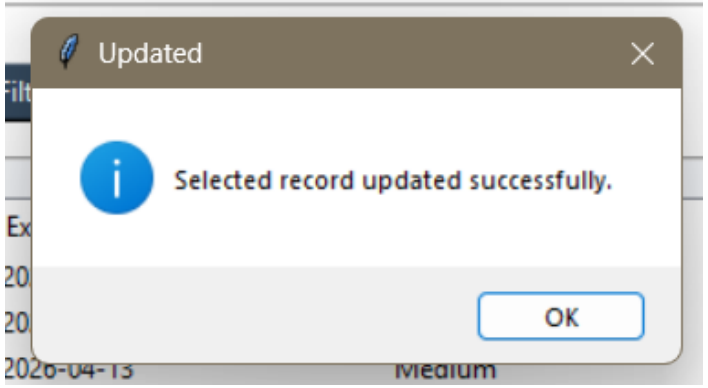
Subject	Topic	Status	Exam Date	Priority	Notes	Days Left	Condition
Science	reproduction	Pending	2026-03-16	Medium		-25	Overdue
maths	algebra	Revising	2026-04-09	Low	2 sums pending	-1	Overdue
Computer Networks	MAC	Revising	2026-04-13	Medium	2 mods pending	3	Urgent
Operating systems	deadlocks	Done	2026-04-15	Medium	1 topic	5	Normal
Math	Calculus	Revising	2026-04-19	Medium		9	Normal
DSA	queues	Done	2026-04-28	High	completed	18	Normal

USER INTERFACE – PLANNER TAB

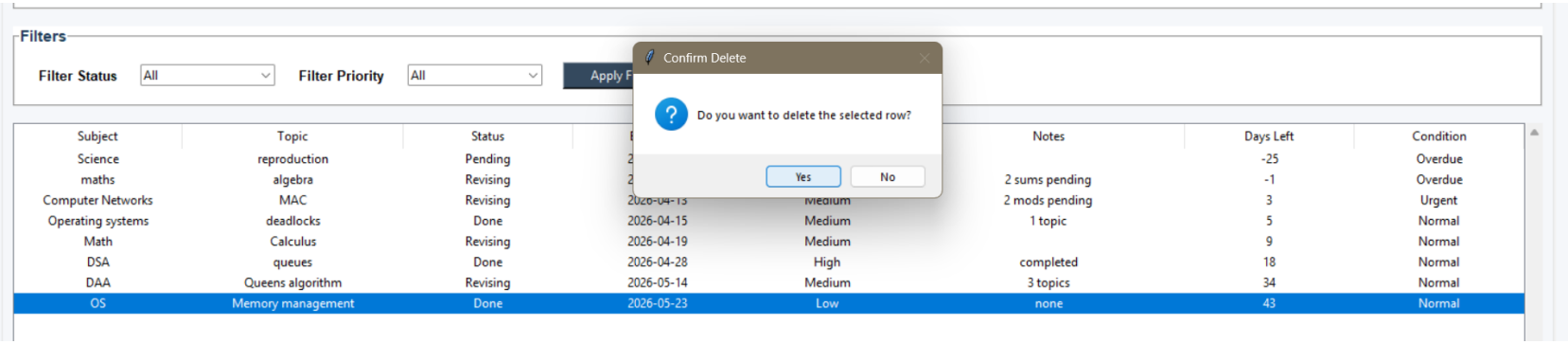
Features:

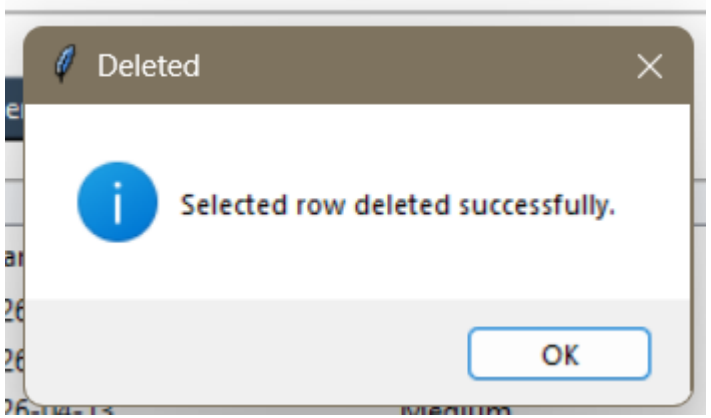
- Input fields for subject & topic
- Status & priority selection
- Table view (Treeview)
- Sorting & filtering

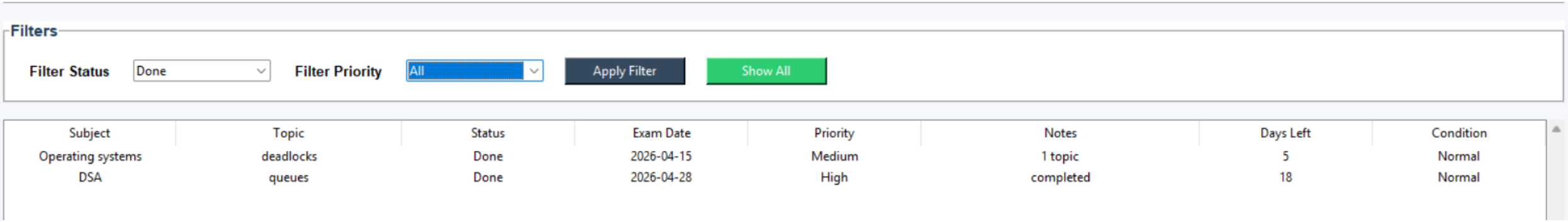




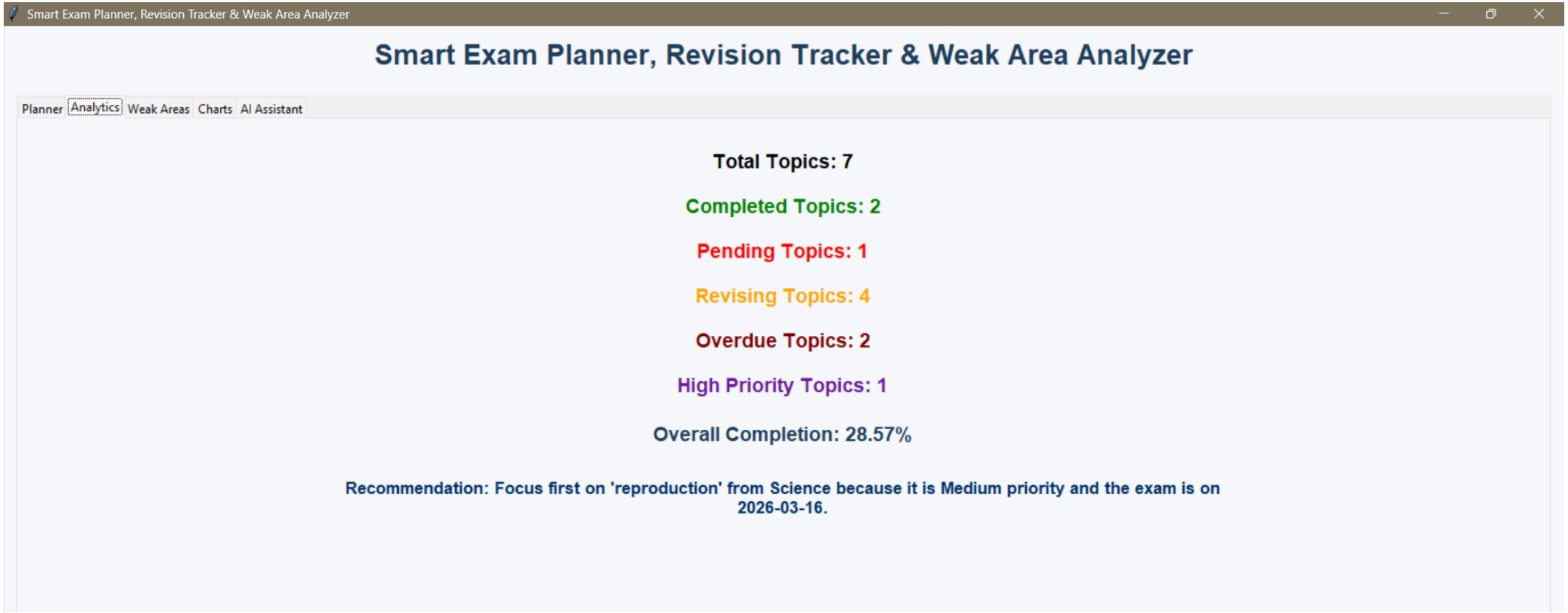
FEATURES


Filters
Filter Status
All
Filter Priority
All
Confirm Delete
Do you want to delete the selected row?
Yes
No
Subject | Topic | Status | Notes | Days Left | Condition
Science | reproduction | Pending | -25 | Overdue
maths | algebra | Revising | 2 sums pending | -1 | Overdue
Computer Networks | MAC | Revising | 2 mods pending | 3 | Urgent
Operating systems | deadlocks | Done | 2026-04-15 | Medium | 1 topic | 5 | Normal
Math | Calculus | Revising | 2026-04-19 | Medium | 9 | Normal
DSA | queues | Done | 2026-04-28 | High | completed | 18 | Normal
DAA | Queens algorithm | Revising | 2026-05-14 | Medium | 3 topics | 34 | Normal
OS | Memory management | Done | 2026-05-23 | Low | none | 43 | Normal


Deleted
Selected row deleted successfully.
OK


Filters
Filter Status
Done
Filter Priority
All
Apply Filter
Show All
Subject | Topic | Status | Exam Date | Priority | Notes | Days Left | Condition
Operating systems | deadlocks | Done | 2026-04-15 | Medium | 1 topic | 5 | Normal
DSA | queues | Done | 2026-04-28 | High | completed | 18 | Normal

ANALYTICS DASHBOARD

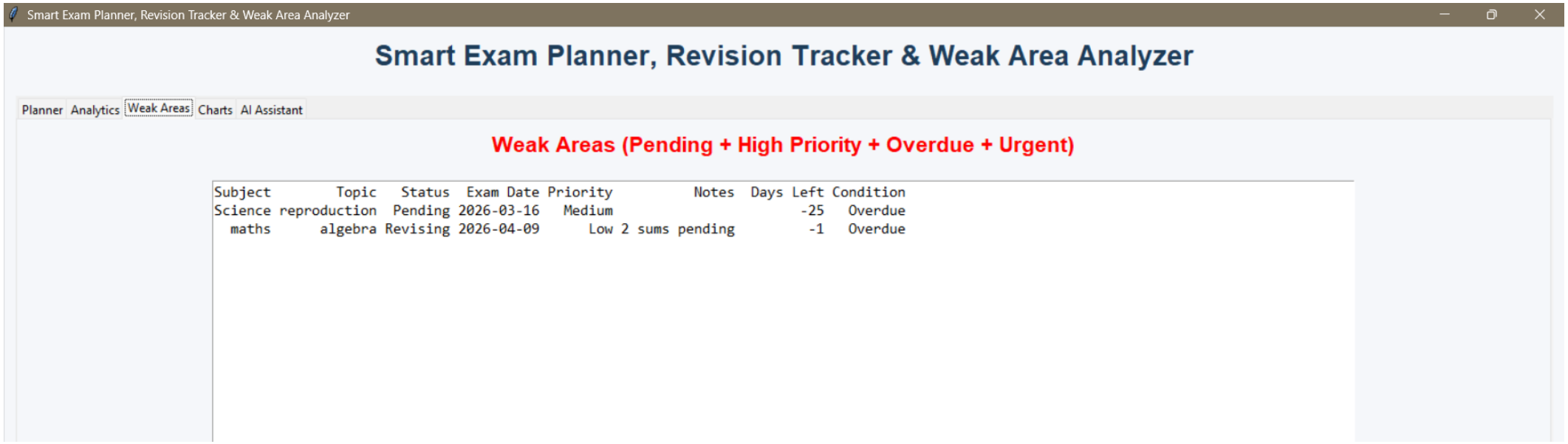


ANALYTICS DASHBOARD

Displays:

- Total topics
- Completed topics
- Pending topics
- Overdue topics
- Completion percentage

WEAK AREA ANALYZER



WEAK AREA ANALYZER

Weak areas detected based on:

- Pending + High Priority
- Overdue topics
- Urgent topics (≤ 3 days left)

SMART RECOMMENDATION ENGINE

Subject	Topic	Status	Exam Date	Priority	Notes	Days Left	Condition
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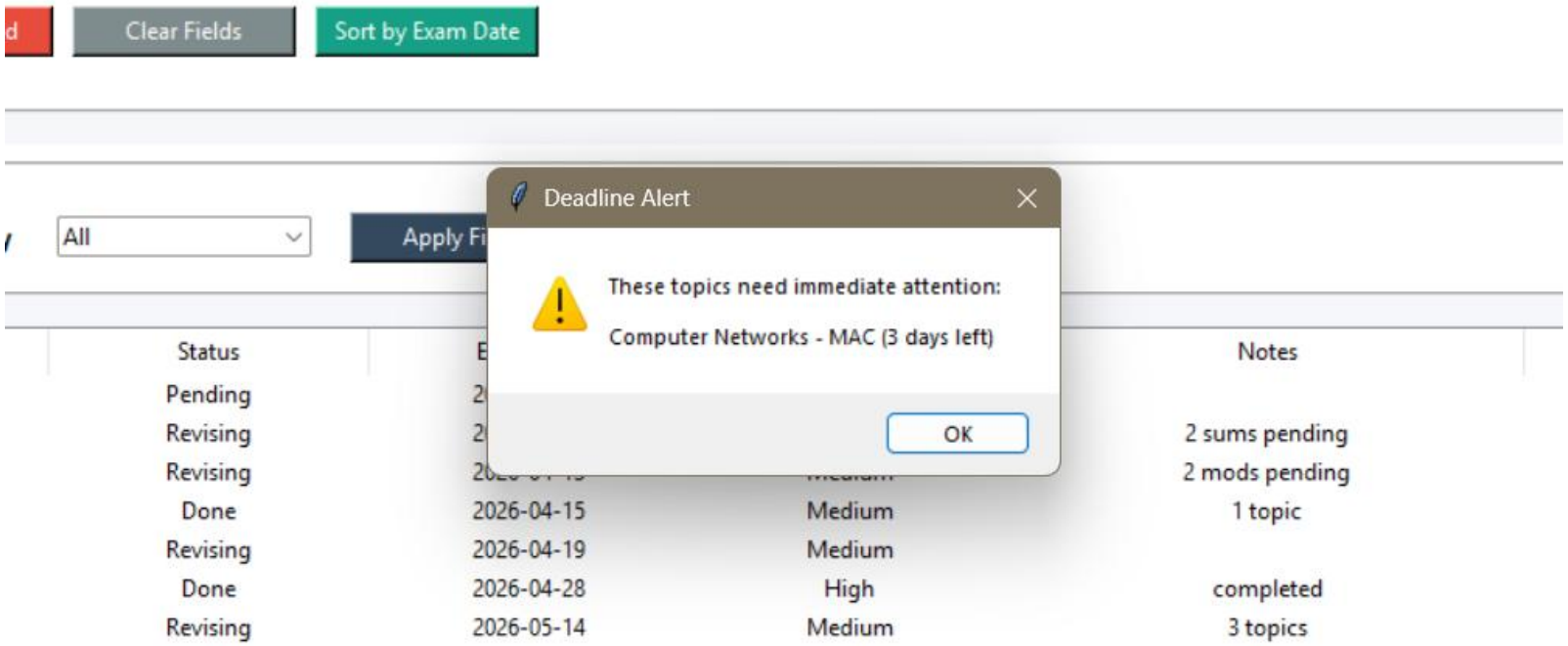
DAA	Queens algorithm	Revising	2026-05-14	Medium	3 topics	34	Normal

SMART RECOMMENDATION ENGINE

- Suggests next topic to study
- Based on:
- Priority
- Exam date
- Helps in efficient preparation

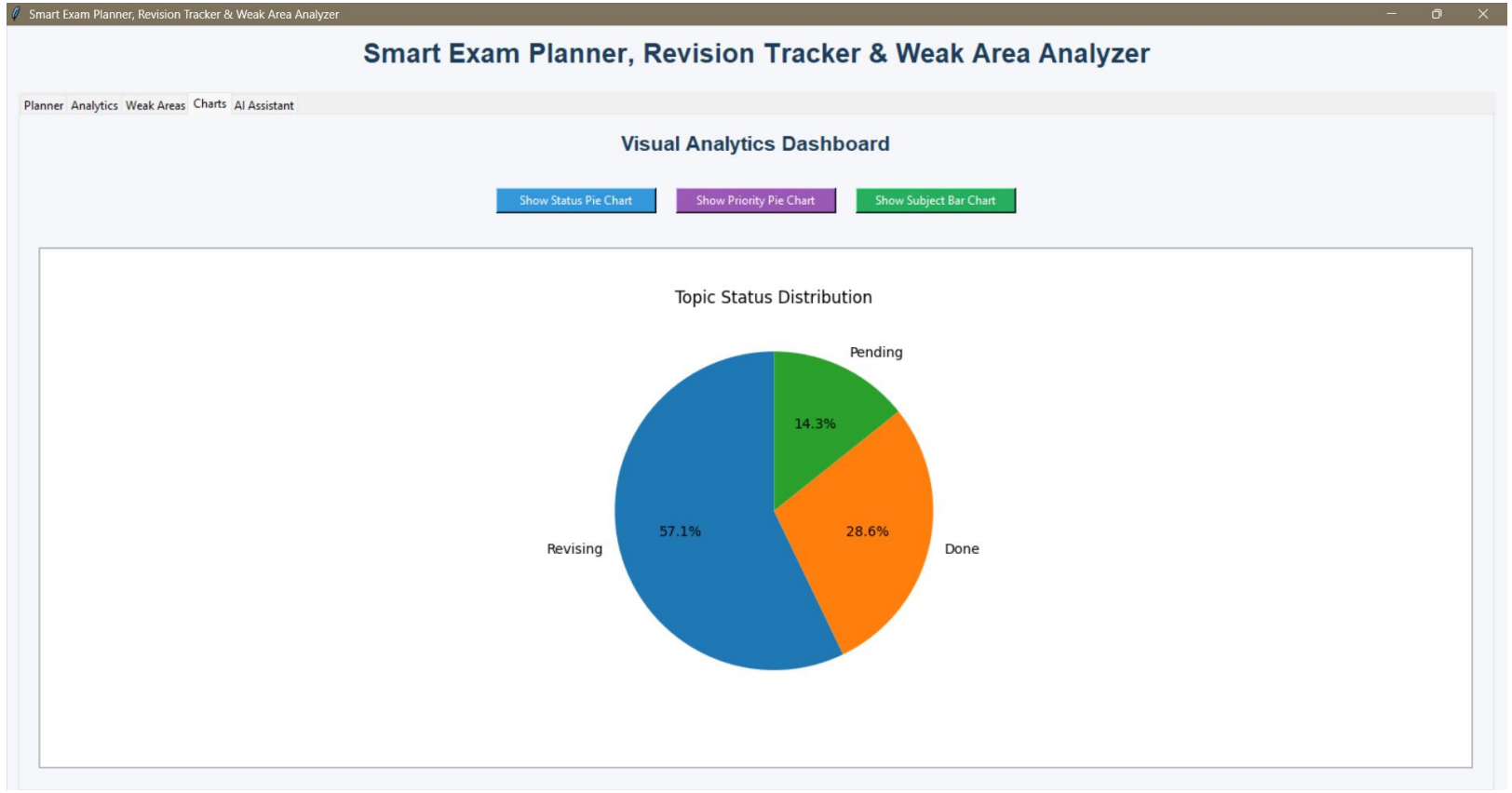
DEADLINE ALERT SYSTEM



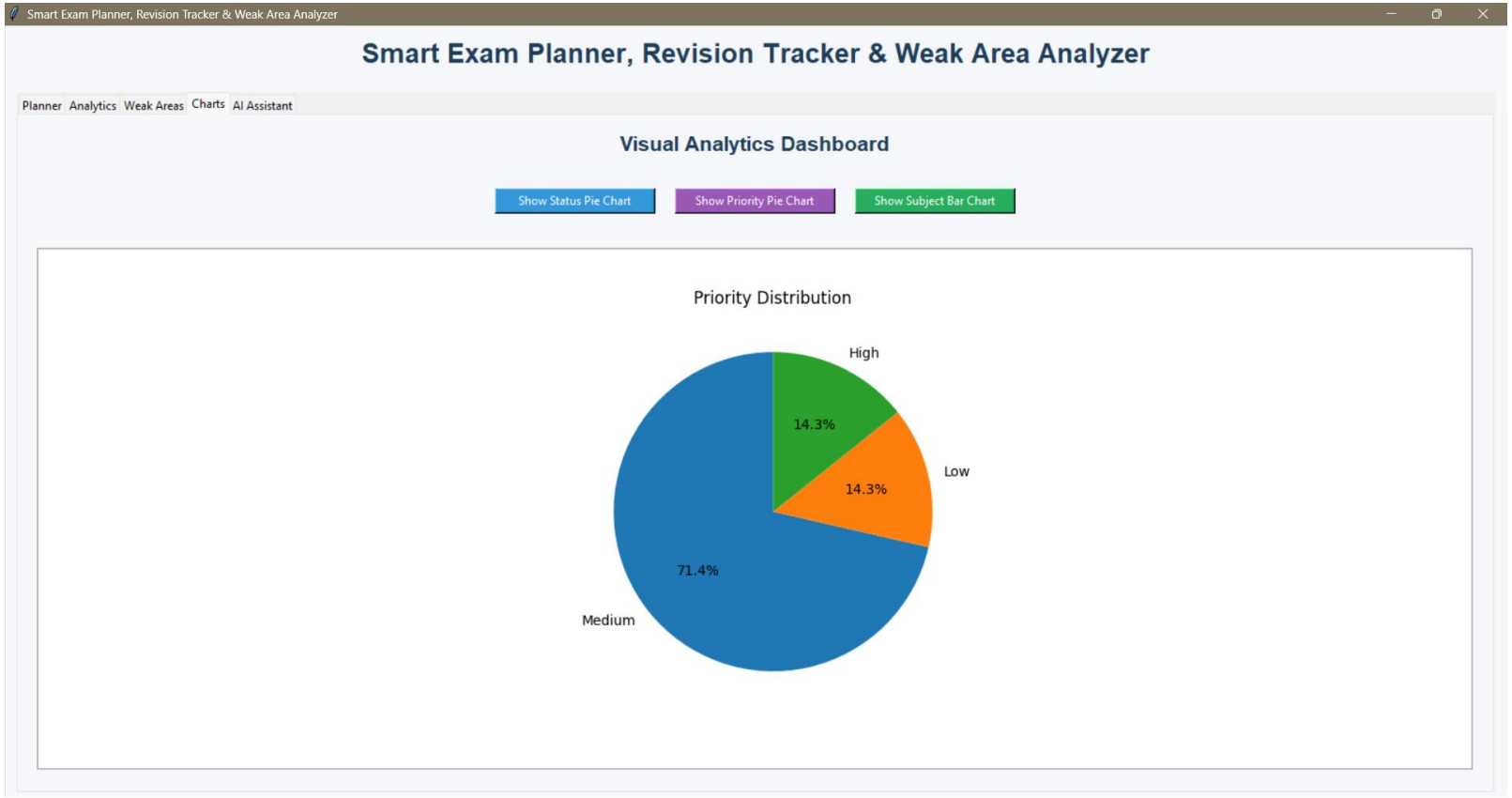
DEADLINE ALERT SYSTEM

- Detects urgent topics
- Displays alert messages
- Ensures timely revision

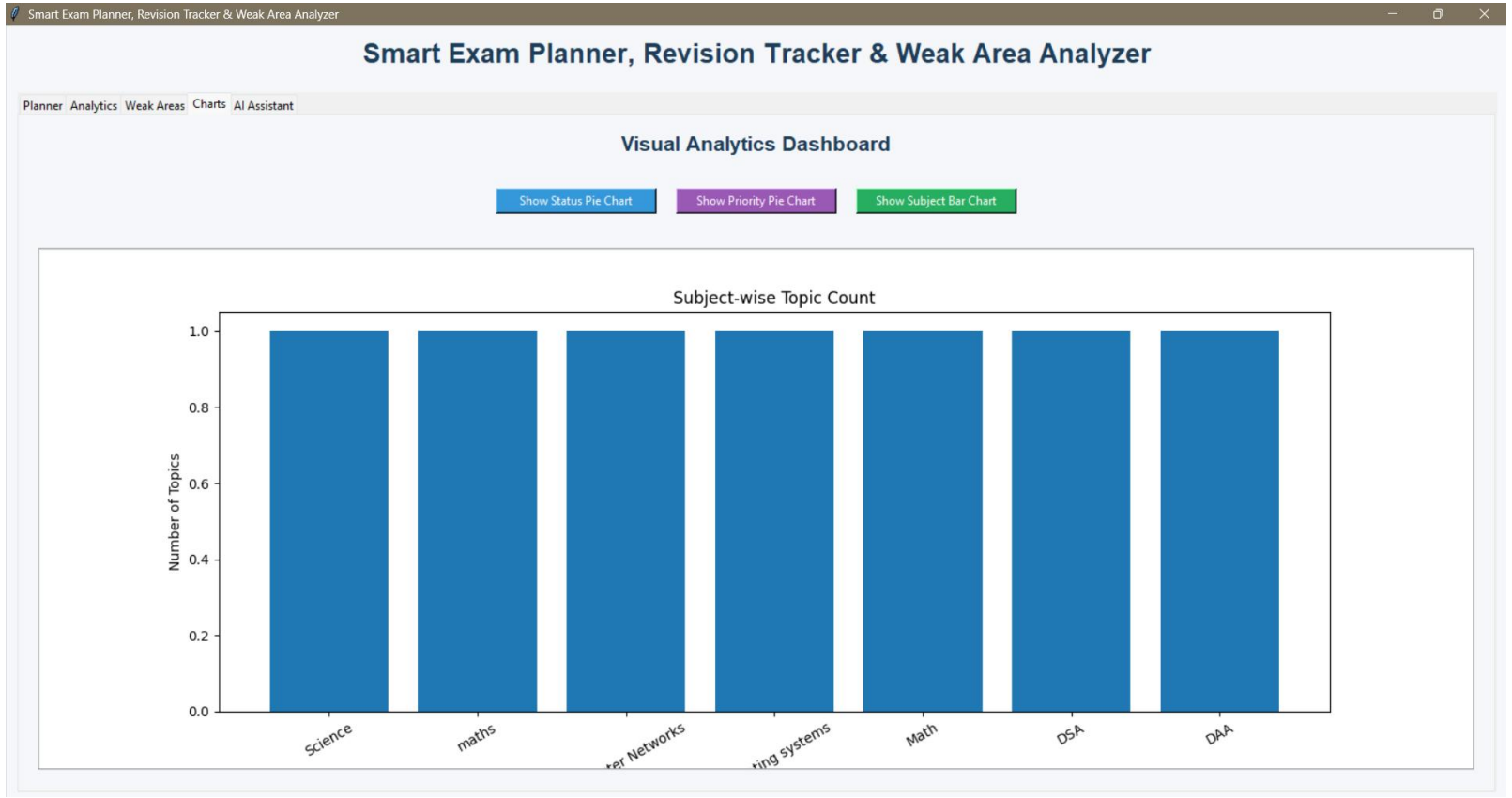
VISUAL ANALYTICS DASHBOARD - Status Pie Chart



Priority Pie Chart



Subject-wise Bar Chart

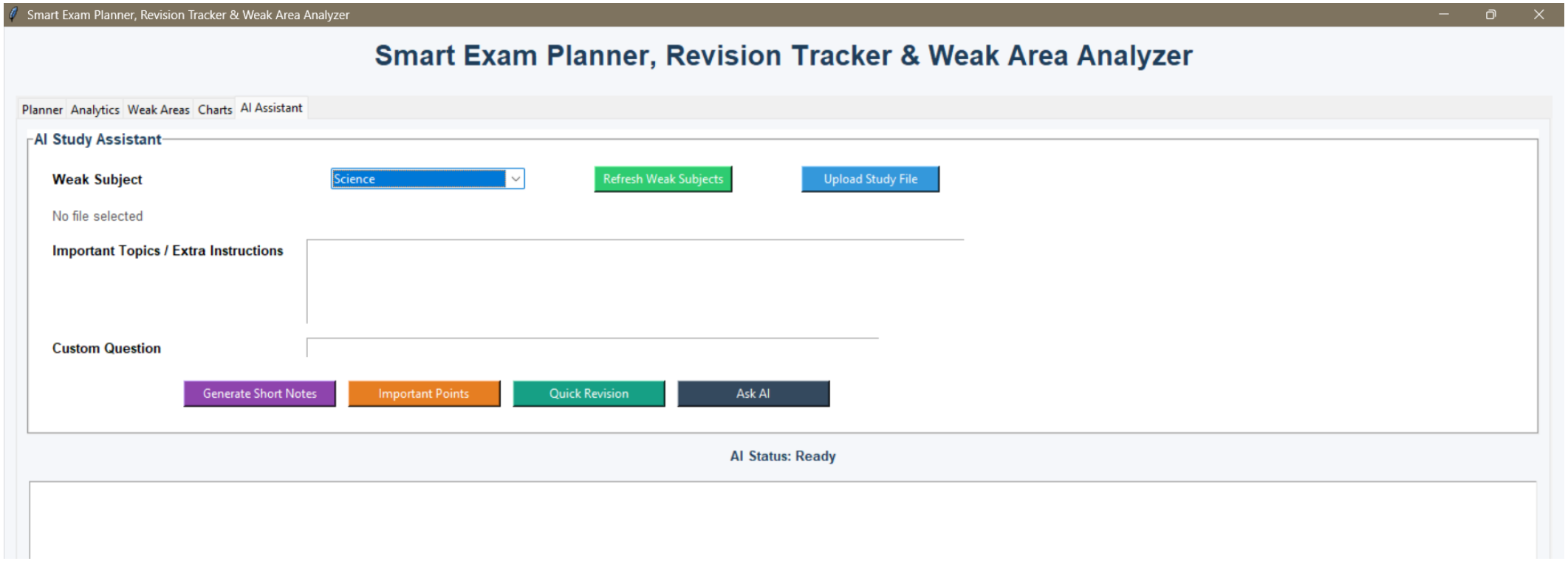


VISUAL ANALYTICS DASHBOARD

Includes:

- Status Pie Chart
- Priority Pie Chart
- Subject-wise Bar Chart

AI STUDY ASSISTANT



AI STUDY ASSISTANT

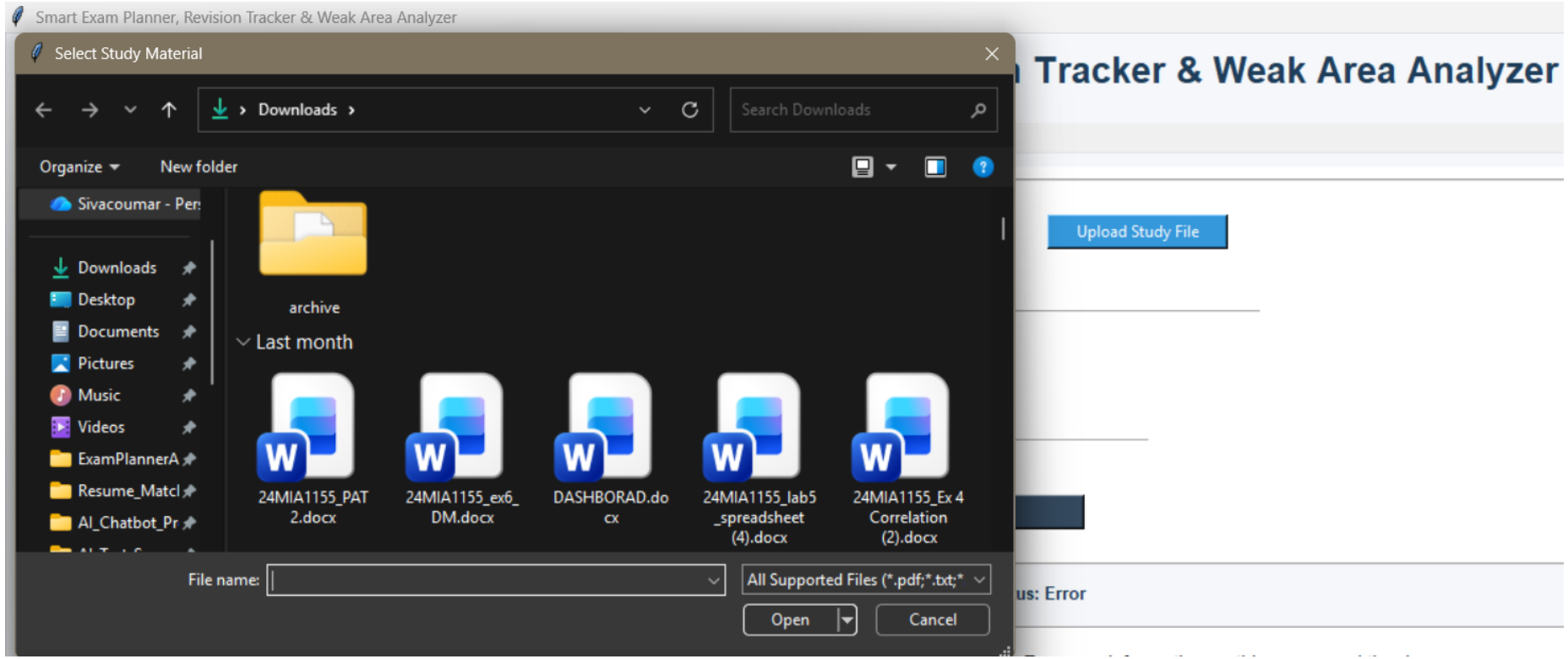
Features:

- Select weak subject
- Upload study material
- Enter custom topics

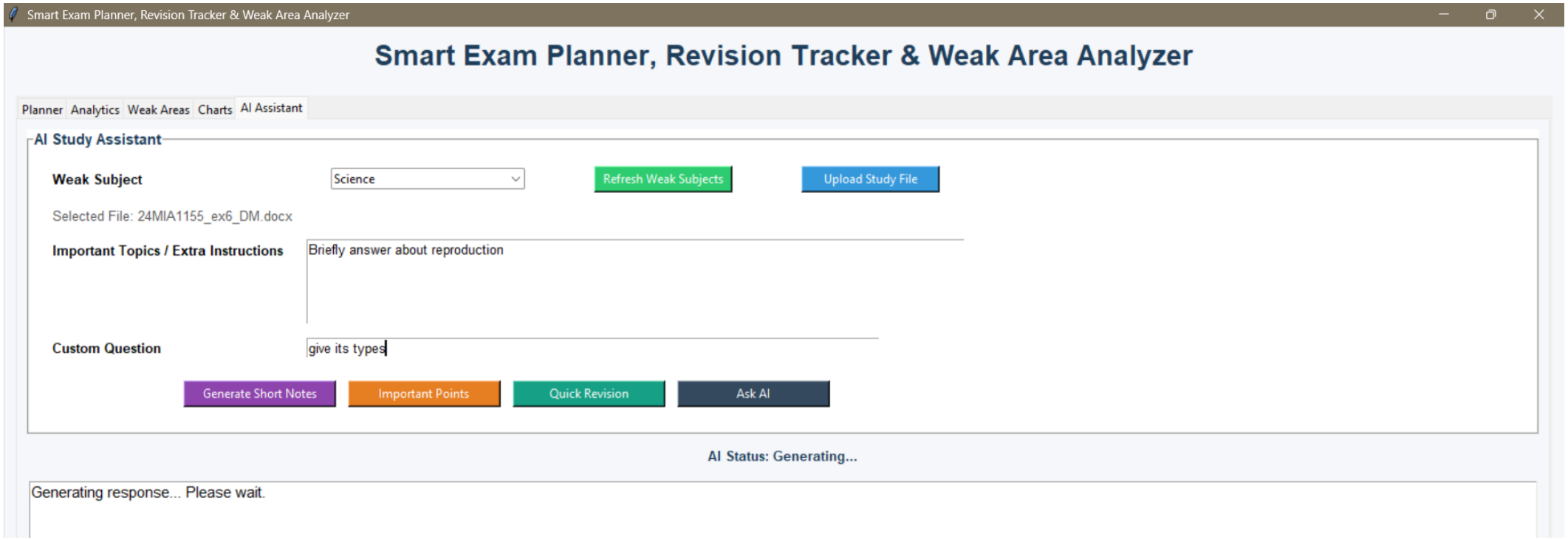
Generate:

- Short notes
- Important points
- Quick revision
- Custom answers

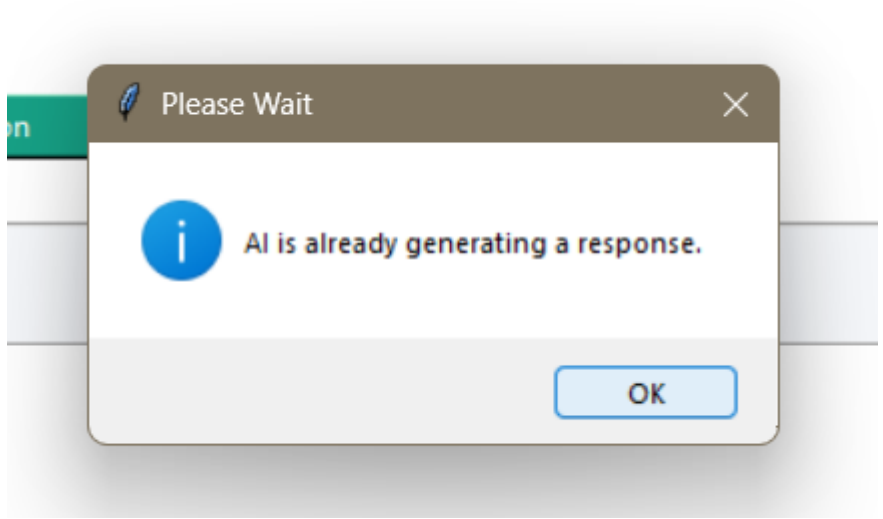
AI WORKING – Upload file



AI WORKING – Ask AI using prompts



AI WORKING



AI WORKING

Flow:

Weak Area → Upload File → AI Processing → Summary Output

- Converts long content → short notes
- Generates exam-focused answers
- Works like intelligent assistant

SPECIAL FEATURES

- Multi-tab interface
- Real-time updates
- Thread-based AI processing
- Smart filtering system
- Error handling & validation

ADVANTAGES

- Improves study efficiency
- Saves time
- Identifies weak areas
- Provides smart suggestions
- Easy to use

LIMITATIONS

- Requires internet for AI
- Depends on API key
- Excel not scalable for large data

FUTURE ENHANCEMENTS

- Mobile app version
- Database integration
- Notification system
- Cloud storage
- Offline AI support

DEMO FLOW

1. Add topic
2. Show analytics
3. Show weak areas
4. Show recommendation
5. Show charts
6. Open AI Assistant
7. Generate notes

CONCLUSION

- Developed a **smart and user-friendly study planner** using Python
- Successfully integrated **data tracking, analytics, and visualization**
- Implemented **automatic weak area detection** and prioritization
- Added **AI-powered assistant** for generating concise study notes
- Enabled **better time management and exam preparation**
- Used Excel effectively as an **intermediary data storage system**
- Transformed a simple tracker into an **intelligent decision-support tool**

This project enhances traditional study planning by combining analytics and AI to support smarter and more efficient learning.